

LANGENOIS, Austria

WMO: 110750

Lat: 48.472N

Lon: 15.697E

Elev: 685

StdP: 14.34

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.4	17.1	6.5	7.9	13.7	11.3	10.0	18.8	19.0	43.4	16.3	42.5	4.9	330	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	89.0	70.6	85.6	69.1	82.5	67.3	72.2	85.4	70.4	82.7	68.7	79.8	5.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	104.9	78.5	66.2	99.0	76.0	64.6	93.6	74.6	36.3	85.8	34.7	83.2	33.3	79.7	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	13.3	11.2	DB	7.3	94.6	7.1	3.4	2.3	97.1	-1.9	99.1	-5.8	101.0	-11.0	103.5	(4)
(5)			WB	6.7	75.3	6.6	1.5	1.9	76.3	-1.9	77.2	-5.6	78.0	-10.4	79.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	31.6	34.8	42.1	51.5	59.4	65.9	69.1	68.4	59.6	50.2	41.5	33.4	(6)	
	DBStd	14.93	7.78	7.93	7.38	7.40	6.22	6.42	5.74	5.94	5.90	6.68	6.67	7.24	(7)	
	HDD50	2193	571	425	260	71	4	0	0	0	2	82	263	515	(8)	
	HDD65	5635	1035	844	711	405	192	67	26	32	181	457	705	980	(9)	
	CDD50	2460	1	1	14	117	297	477	593	572	290	90	8	0	(10)	
	CDD65	428	0	0	0	2	20	94	154	139	19	0	0	0	(11)	
	CDH74	4018	0	0	0	46	233	897	1437	1246	154	5	0	0	(12)	
	CDH80	1243	0	0	0	6	33	264	493	425	22	0	0	0	(13)	
(14) Wind	WSAvg	4.6	4.6	4.9	5.6	5.5	5.0	4.8	4.8	4.1	4.3	4.1	3.9	3.9	(14)	
(15) Precipitation	PrecAvg	24.1	1.1	0.9	1.5	1.5	2.9	3.5	3.3	3.2	2.2	1.7	1.3	1.1	(15)	
	PrecMax	34.4	2.4	2.0	3.8	3.6	5.7	9.3	6.9	10.1	5.7	3.7	2.8	2.1	(16)	
	PrecMin	15.8	0.1	0.1	0.2	0.0	0.7	1.7	1.1	0.7	0.6	0.2	0.0	0.2	(17)	
	PrecStd	4.3	0.6	0.6	0.9	0.9	1.4	1.6	1.5	1.9	1.1	1.0	0.6	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	59.1	68.1	79.4	83.7	90.7	93.0	93.5	82.9	72.2	63.6	54.2	(19)	
		MCWB	47.3	48.4	53.5	60.3	65.9	70.9	71.3	70.7	66.6	60.1	54.3	47.9	(20)	
	2%	DB	50.4	54.1	63.0	73.5	79.7	86.3	88.9	88.8	78.4	67.5	58.1	49.7	(21)	
		MCWB	44.3	46.4	51.1	58.3	64.6	70.1	70.9	70.1	64.9	58.3	51.7	44.5	(22)	
	5%	DB	46.4	49.9	58.2	69.3	76.4	82.8	85.5	84.6	74.4	63.9	54.0	45.8	(23)	
		MCWB	41.3	43.5	48.8	56.0	62.7	67.5	69.0	68.7	63.1	56.8	49.7	41.6	(24)	
	10%	DB	42.8	46.0	54.0	65.5	72.7	79.4	82.0	81.1	70.6	60.5	50.7	43.0	(25)	
		MCWB	38.7	40.9	46.4	54.1	60.3	66.0	67.5	67.0	61.0	55.0	47.0	39.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	49.4	54.6	62.2	68.1	73.7	74.8	74.5	68.2	62.3	55.5	48.5	(27)	
		MCDB	54.9	57.1	66.0	76.6	80.6	87.1	88.6	87.7	79.4	68.8	61.1	53.4	(28)	
	2%	WB	44.6	46.5	51.8	59.2	65.6	71.3	72.2	71.7	66.1	59.8	52.8	44.8	(29)	
		MCDB	50.3	53.5	61.3	72.0	77.3	83.6	85.6	85.1	76.1	65.5	57.5	49.0	(30)	
	5%	WB	41.7	43.9	49.5	56.9	63.7	69.2	70.3	69.9	64.2	57.8	50.0	42.3	(31)	
		MCDB	45.7	49.5	56.8	67.4	74.2	80.0	82.5	82.5	72.6	62.5	53.6	45.2	(32)	
	10%	WB	39.2	41.2	47.2	54.8	61.7	67.2	68.6	68.0	62.2	55.8	47.6	40.1	(33)	
		MCDB	42.6	45.4	53.4	64.2	70.7	77.0	79.5	78.5	68.9	60.0	50.3	42.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	13.2	16.4	19.7	19.3	19.5	19.9	20.3	17.6	15.1	10.6	9.7	(35)	
		MCDBR	16.0	20.5	25.1	26.7	25.9	25.6	26.2	26.4	23.5	20.5	16.9	15.8	(36)	
	5% WB	MCWBR	11.2	13.2	14.3	13.1	11.8	11.0	10.4	10.9	11.2	12.0	11.1	11.3	(37)	
		MCWBR	15.3	19.8	23.4	24.7	23.8	23.6	24.8	21.9	18.3	15.8	15.8	15.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.311	0.335	0.376	0.419	0.420	0.425	0.432	0.429	0.400	0.369	0.341	0.310	(40)	
		taud	2.386	2.348	2.279	2.193	2.232	2.255	2.237	2.262	2.310	2.373	2.405	2.423	(41)	
	Ebn at Noon	Ebn at Noon	239	258	264	263	267	266	262	257	253	241	224	224	(42)	
		Edn at Noon	22	28	35	43	43	42	42	39	34	27	21	19	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	313	590	932	1377	1597	1745	1683	1502	1072	662	337	247	(44)	
	RadStd	RadStd	48	78	121	162	183	150	163	150	138	77	35	27	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (92 neighbors)		+0.94	N/A	N/A	N/A	+0.70	+0.59	-216	-272	+131	+59	

Nomenclature: See separate page